



YEAR 12 EXTENSION 2 MATHEMATICS JUNE 2007

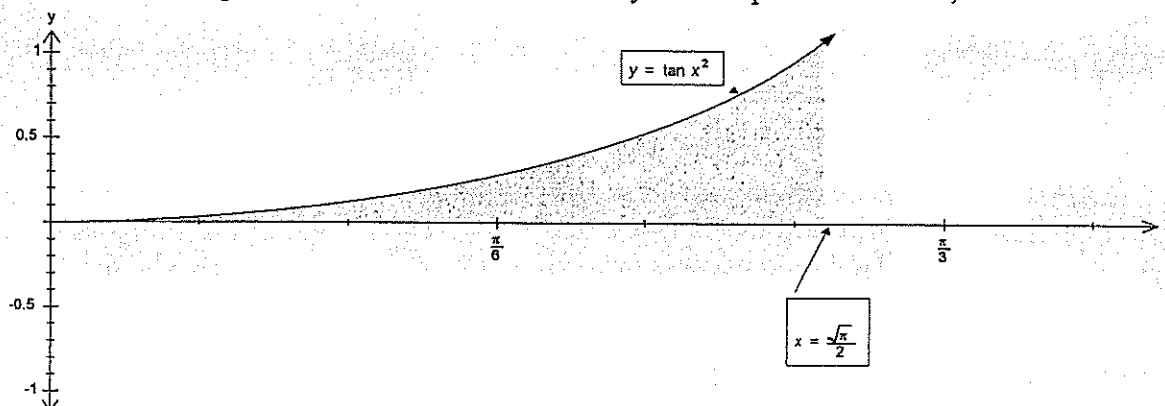
NAME		RESULT	
DIRECTIONS	- Full working should be shown in every question. Marks may be deducted for careless or badly arranged work. - Use black or blue pen only (<i>not pencils</i>) to write your solutions. - No liquid paper is to be used. If a correction is to be made, one line is to be ruled through the incorrect answer.		

TIME ALLOWED: 70 MINUTES

- Find $\int \frac{4x - 6}{(x+1)(2x^2 + 3)} dx$ 3
- Find $\int (\ln x)^2 dx$ by using the technique of integration by parts 3
- Find $\int \frac{x^2 dx}{\sqrt{4-x^2}}$ 4
- Evaluate $\int_0^{\frac{\pi}{2}} \frac{dx}{5+4\cos x}$ by using the substitution $t = \tan \frac{x}{2}$. 4
- Find $\int \frac{dx}{\sqrt{7-2x-x^2}}$ 2
- Use the result $\int_0^a f(x) dx = \int_0^a f(a-x) dx$ to show that $\int_0^{\frac{\pi}{4}} \frac{\cot(x + \frac{\pi}{4}) dx}{1 - \sin 2x} = \frac{1}{2} \int_0^{\frac{\pi}{4}} \sec^2 x \tan x dx$. 3
 - Hence show that $\int_0^{\frac{\pi}{4}} \frac{\cot(x + \frac{\pi}{4}) dx}{1 - \sin 2x} = \frac{1}{4}$ 2
- Use a suitable substitution to prove that $\int_{-1}^0 x(1+x)^n dx = \frac{-1}{(n+1)(n+2)}$ 3

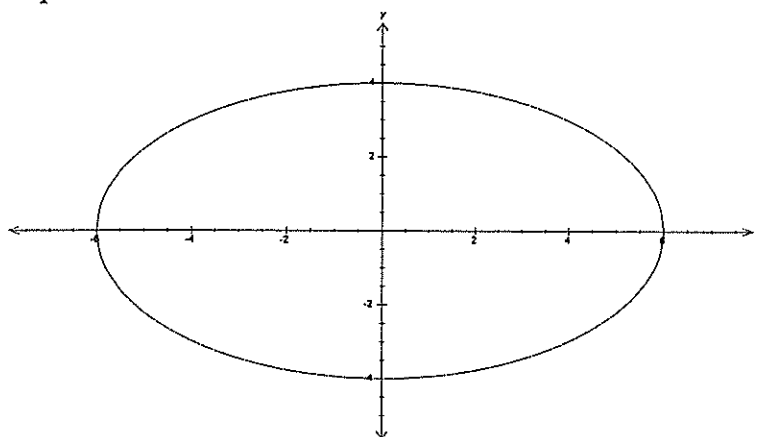
8. i. Find $\frac{d}{dx}(\ln(\cos x^2))$ 1

- ii. The region in the plane bounded by the curve $y = \tan x^2$, the x axis and the line $x = \frac{\sqrt{\pi}}{2}$ 4
shown in the diagram below is rotated about the y axis to produce a solid, S.



Use the method of cylindrical shells to show the volume of S is $\frac{\pi}{2} \ln 2$ units³.

9. A solid has an elliptical base with equation $\frac{x^2}{36} + \frac{y^2}{16} = 1$. Sections of the solid perpendicular to its base and parallel to the minor axis are semicircles.



- i. Show that the area of a slice of the solid is given by $A(x) = \frac{2\pi}{9}(36 - x^2)$ units². 2
- ii. Hence, find the volume of the solid. 2
10. The area between the parabola $y = 6x - x^2$ and the x axis is rotated about the y axis. When the area is rotated a line segment, L, at height y sweeps out an annulus. The x ordinates of the end points of L are x_1 and x_2 .
- i. Show that the area of the annulus is equal to $A(y) = 12\pi\sqrt{9 - y}$ units². 2
- ii. Hence find the volume of the solid formed. 3
11. $I_n = \int_0^1 x^n \sqrt{1-x} \, dx$. Show that for $n > 0$, $I_n = \left(\frac{2n}{2n+3}\right)I_{n-1}$, where n is an integer. 4

EXTENSION 2 JUNE 2007

$$1. \text{ let } \int \frac{4x-6}{(x+1)(2x^2+3)} dx = \int \left(\frac{A}{x+1} + \frac{Bx+C}{2x^2+3} \right) dx \quad \textcircled{1} \text{ for either}$$

$$4x-6 = A(2x^2+3) + (Bx+C)(x+1)$$

$$\text{Put } x = -1$$

$$-10 = 5A + 0$$

$$A = -2$$

Equating coefficients:

$$\text{of } x^2: 0 = 2A + B$$

$$B = 2 \times 2$$

$$B = 4$$

$$\text{of } x^0: -6 = 3A + C$$

$$C = 0$$

$$A = -2, B = 4, C = 0$$

$$\textcircled{1} \text{ or CFP } A = -2, B = 0$$

$$\therefore \int \frac{4x-6}{(x+1)(2x^2+3)} dx = \int \frac{-2}{x+1} + \frac{4x}{2x^2+3} dx$$

$$= -2 \ln(x+1) + \ln(2x^2+3) + C \quad \textcircled{1} \text{ no CFP if B omitted above}$$

$$2. \int (\ln x)^2 dx = x(\ln x)^2 - \int 2 \ln x dx$$

$$u = (\ln x)^2 \quad v' = 1$$

$$u' = 2 \ln x \quad v = x$$

$$= x(\ln x)^2 - 2 \int \ln x dx$$

$$u = \ln x \quad v' = 1$$

$$u' = \frac{1}{x} \quad v = x$$

$$= x(\ln x)^2 - 2 \left(x \ln x - \int 1 dx \right)$$

$$= x(\ln x)^2 - 2x \ln x + 2x + C$$

$$3. \int \frac{x}{\sqrt{4-x^2}} dx = \int \frac{4 \sin^2 \theta \cdot 2 \cos \theta d\theta}{\sqrt{4(1-\sin^2 \theta)}} \quad \textcircled{1} \text{ let } x = 2 \sin \theta$$

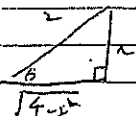
$$dx = 2 \cos \theta d\theta \quad \textcircled{1}$$

$$= \int \frac{4 \sin^2 \theta \cos \theta d\theta}{2 \cos \theta}$$

$$= \int 2(1 - \cos^2 \theta) d\theta$$

$$= 2\theta - \sin 2\theta + C \quad \textcircled{1}$$

$$= 2 \sin^{-1} \frac{x}{2} - 2 \cdot \frac{x}{2} \sqrt{4-x^2} + C = 2 \sin^{-1} \frac{x}{2} - x \sqrt{4-x^2} + C \quad \textcircled{1}$$



$$4. \int_0^{\pi/4} \frac{dx}{5+9 \cos^2 x} = \int_0^1 \frac{2 dt}{(t^2+1)(5+4(\frac{1-t^2}{1+t^2}))} \quad \textcircled{1}$$

$$= \int_0^1 \frac{2 dt}{5t^2+5+4-4t^2}$$

$$= \int_0^1 \frac{2 dt}{t^2+9}$$

$$= \frac{2}{3} \left[\tan^{-1} \frac{t}{3} \right]_0^1 \quad \textcircled{1}$$

$$= \frac{2}{3} \left(\tan^{-1} \frac{1}{3} - \tan^{-1} 0 \right)$$

$$= \frac{2}{3} \tan^{-1} \frac{1}{3} \quad \textcircled{1}$$

$$\text{let } t = \tan \frac{x}{2}$$

$$\frac{dx}{dt} = \frac{2}{1+t^2}$$

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$$\text{when } x = \frac{\pi}{4}, t = \tan \frac{\pi}{4} = 1$$

$$\text{when } x = 0, t = \tan 0 = 0 \quad \textcircled{1}$$

$$5. \int \frac{dx}{\sqrt{7-2x-x^2}} = \int \frac{dx}{\sqrt{8-(x+1)^2}}$$

$$7-2x-x^2 = 7-(x^2+2x+1)+1 = 8-(x+1)^2 \quad \textcircled{1}$$

$$= \sin^{-1} \left(\frac{x+1}{2\sqrt{2}} \right) + C \quad \textcircled{1} \text{ (or equivalent expression to } \sqrt{2} \text{)}$$

$$6a) \int_0^{\pi/4} \frac{\cot(x+\pi/4) dx}{1+\sin 2x} = \int_0^{\pi/4} \frac{\cot(\pi/4+x+\pi/4) dx}{1+\sin(2(\pi/4-x))} \quad \textcircled{1}$$

$$= \int_0^{\pi/4} \frac{\cot(\pi/2-x)}{1+\sin(\pi/2-2x)} dx$$

$$= \int_0^{\pi/4} \frac{\tan x dx}{1+\cos 2x}$$

$$= \int_0^{\pi/4} \frac{\tan x dx}{2 \cos^2 x} = \frac{1}{2} \int_0^{\pi/4} \tan x \sec^2 x dx \quad \textcircled{1}$$

$$b) \int_0^{\pi/4} \tan x \sec^2 x dx = \left[\frac{1}{2} \tan^2 x \right]_0^{\pi/4} \quad \textcircled{1}$$

$$= \frac{1}{2} (1 - 0) = \frac{1}{2} \quad \textcircled{1}$$

7. Let $u = 1+x$ $du = dx$
 when $x=0$, $u=1$
 $x=-1$, $u=0$

$$\int_{-1}^0 x(1+x)^n dx = \int_1^0 (u-1)u^n du \quad (1) \text{ including limits}$$

$$= \int_1^0 u^{n+1} - u^n du$$

$$= \left[\frac{u^{n+2}}{n+2} - \frac{u^{n+1}}{n+1} \right]_1^0$$

$$= \frac{1}{n+2} - \frac{1}{n+1}$$

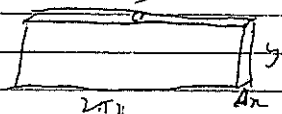
$$= \frac{n+1 - (n+2)}{(n+1)(n+2)}$$

$$= \frac{-1}{(n+1)(n+2)}$$

$$\therefore \int_{-1}^0 x(1+x)^n dx = \frac{-1}{(n+1)(n+2)}$$

8. (i) $\frac{d}{dx} (\ln(\cos x^2)) = \frac{-2x \sin x^2}{\cos x^2}$
 $= -2x \tan x^2$

(ii) ~~Volume of shell~~ $= \int_0^{\sqrt{36}} 2\pi x y dx$



Volume of shell, $\Delta V = 2\pi x y \Delta x$
 $= 2\pi x \tan x^2 \Delta x$

Volume of solid, $V = \lim_{\Delta x \rightarrow 0} \sum_{n=0}^{\sqrt{36}} 2\pi x \tan x^2 \Delta x$

$$= 2\pi \int_0^{\sqrt{36}} x \tan x^2 dx \quad (1)$$

$$= -\pi \left[\ln(\cos x^2) \right]_0^{\sqrt{36}} \quad (1)$$

$$= -\pi (\ln(\cos 36) - \ln(\cos 0))$$

$$= -\pi \left(\ln \frac{1}{2} - \ln 1 \right) \quad (1) \quad [4]$$

$$= -\pi (\ln 2 - 0)$$

$V = \frac{\pi}{2} \ln 2 \text{ units}^3$ as reqd. $(1) \text{ ignore units}$

9. (i) Area of disc $= \frac{\pi y^2}{2} \quad (1)$

$$= \frac{\pi}{2} \left(16 \left(1 - \frac{x^2}{36} \right) \right)$$

$$= \frac{16\pi}{2} \left(\frac{36-x^2}{36} \right)$$

$$= \frac{2\pi}{9} (36-x^2) \text{ units}^2 \quad (1) \text{ ignore units}$$

(ii) Volume of disc $\Delta V = \frac{2\pi (36-x^2)}{9} \Delta x$

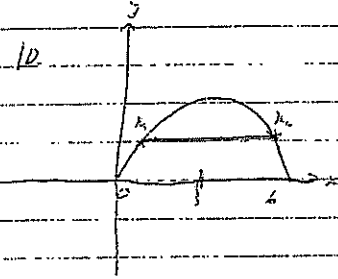
Volume of solid $\sum_{n=0}^6 \frac{2\pi (36-x^2)}{9} \Delta x$

$$= \frac{4\pi}{9} \int_0^6 (36-x^2) dx \quad (1)$$

$$= \frac{4\pi}{9} \left[36x - \frac{x^3}{3} \right]_0^6$$

$$= \frac{4\pi}{9} \left(216 - \frac{216}{3} - 0 \right)$$

$$= 64\pi \text{ units}^3 \text{ ignore units } (1)$$



(i) $A = \pi x_1^2 - \pi x_2^2$
 $= \pi (x_1 + x_2)(x_1 - x_2)$

Now $x_1^2 - 6x_1 + 9 = 0$

$$1 = \frac{6 \pm \sqrt{36-36}}{2}$$

$$= \frac{6 \pm \sqrt{0}}{2}$$

$$= 3 \pm \sqrt{0}$$

(1)

$$x_1 = 3 + \sqrt{9-y} \quad x_2 = 3 - \sqrt{9-y}$$

$$\therefore \Delta y = \pi (3 + \sqrt{9-y} + 3 - \sqrt{9-y}) (3 + \sqrt{9-y} - 3 + \sqrt{9-y})$$

[2]

$$\Delta y = 12\pi \sqrt{9-y}$$

(1)

10(ii) : Volume of slice, $\Delta V = 12\pi \sqrt{9-y} \Delta y$

$$\text{Volume} = \lim_{\Delta y \rightarrow 0} \sum_{y=0}^9 12\pi \sqrt{9-y} \Delta y$$

when $x=3, y=9$ (1)

$$= 12\pi \int_0^9 (9-y)^{\frac{1}{2}} dy$$

$$= 12\pi \left[\frac{2}{3} (9-y)^{\frac{3}{2}} \right]_0^9$$

(1) [3]

$$= -8\pi (9-9)^{\frac{3}{2}} - (9-0)^{\frac{3}{2}}$$

$$= -8\pi (0 - 27)$$

$$= 216\pi \text{ units}^3$$

(1)

11. $I_n = \int_0^1 x^n \sqrt{1-x} dx$

$$\left. \begin{aligned} u &= x^n & v &= \sqrt{1-x} \\ u' &= nx^{n-1} & v' &= -\frac{1}{2}(1-x)^{-\frac{1}{2}} \end{aligned} \right\} (1)$$

$$= \left[\frac{-2}{3} (1-x)^{\frac{3}{2}} x^n \right]_0^1 - \int_0^1 \frac{-2n}{3} x^{n-1} (1-x)^{\frac{1}{2}} dx$$

(1)

$$= (0-0) + \frac{2n}{3} \int_0^1 (1-x) (x^{n-1} \sqrt{1-x}) dx$$

(1)

$$= \frac{2n}{3} \left(\int_0^1 x^{n-1} \sqrt{1-x} dx - \int_0^1 x^n \sqrt{1-x} dx \right)$$

$$I_n + \frac{2n}{3} I_n = \frac{2n}{3} I_{n-1}$$

$$I_n \left(\frac{3+2n}{3} \right) = \frac{2n}{3} I_{n-1}$$

$$I_n = \frac{2n}{2n+3} I_{n-1}$$

[4]

(1)

11.11.11 - 12.11.11